



K. J. Somaiya School of Engineering

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Experiment / assignment / tutorial No. 1

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of the Staff In-charge with date

S. Somaiya
29/04/26

TITLE: Frequency Response of CE Amplifier

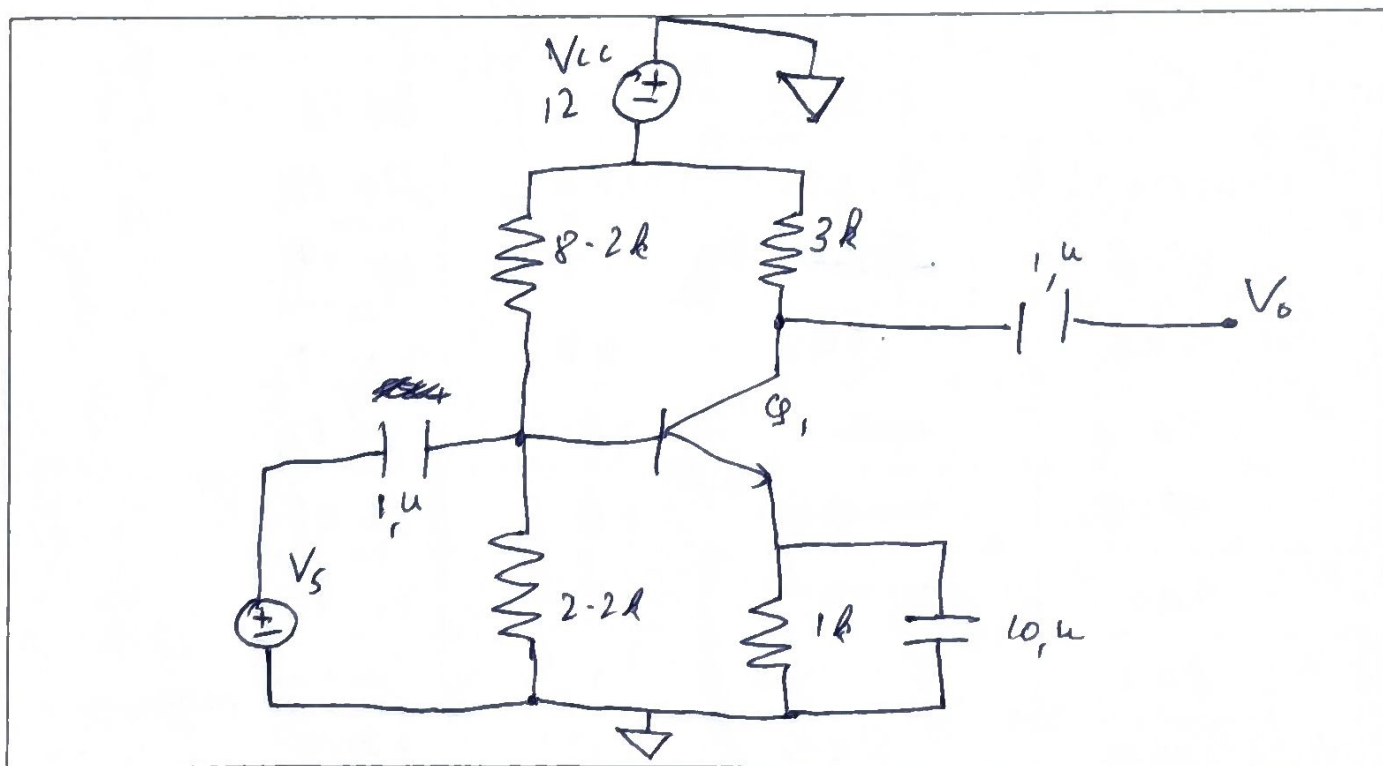
AIM: To calculate bandwidth of single stage CE amplifier

OUTCOME: Students will be able to analyze the frequency response of BJT amplifier.

PROCEDURE:

1. Implement the circuit as per the circuit diagram.
2. Take sinusoidal input signal of fixed amplitude and apply it to the amplifier circuit.
3. Vary the frequency of the source and measure the output voltage.
4. Find frequency response of amplifier circuits with reference to observation table I.
5. Plot the frequency (f) Vs Gain (A_v) on a semi-log Graph paper
6. Draw a horizontal line at 0.707 times A_v and note down the cut off points (lower and higher cut-off frequencies) and the Bandwidth is calculated as $B.W = f_2 - f_1$.

Circuit Diagram:



Observation Table I

$V_{CC} =$ V

Input Signal Voltage (V_s) = V



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Sr. No	Input Signal (Vs) frequency	Output Voltage (Vo)	Voltage Gain (Vo/Vs)	Gain in dB
1.	100	320 mV	16	24.08
2.	200	840 mV	32	30.10
3.	300	980 mV	49	33.8
4.	500	1.31 V	65.5	36.3
5.	1000	1.79 V	89.5	39.03
6.	1.5 k	1.99 V	99.5	39.95
7.	2 k	2.09 V	104.5	39.45 40.38
8.	2.5 k	2.21 V	110.5	40.88
9.	3 k	2.2 V	110	40.802
10.	3.5 k	2.27 V	113.5	41.09
11.	4 k	2.21 V	110.5	40.86
12.	10 k	2.15 V	107.5	40.62
13.	150 k	2.2 V 2.2 V	113.5	41.09
14.	100 k	1.99 V	109.5	40.78
15.	200 k	1.99 V 1.99 V	98.5	39.86
16.	500	1.31 V 1.31 V	66.5	36.06
17.	600	1.23 V 1.23 V	61.5	34.92
18.	700	1.09 V	54.5	34.46

19. 1000 830 mV 41.5 32.36
 20. 1600 620 mV 31 29.8



Observation Table II

	Practical(Ltspice)	Theoretical
Lower Cut off Frequency	0.9 kHz	1.14 kHz
Higher Cut off Frequency	0.430 MHz	0.465 MHz
Bandwidth	0.4281 MHz	0.4453 MHz

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Theoretical Calculations

Voltage gain, Lower and higher cut-off frequencies

$$|A_v|_{mid} = 210$$

$$f_{LCE} = f_L = 1.14 \text{ kHz} \quad f_{Ha} = f_H = 0.4465 \text{ MHz}$$

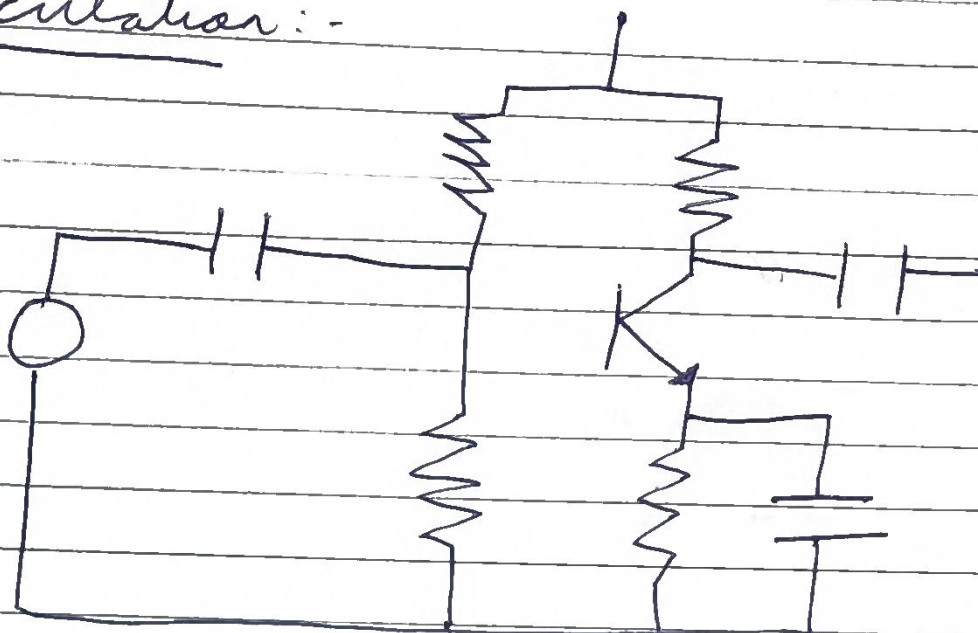
$$B_W = 0.4453 \text{ MHz}$$

Conclusion: In this experiment, the frequency response of a single stage CE amplifier was studied successfully. The variation of the voltage gain with frequency was observed.



AE Experiment 01

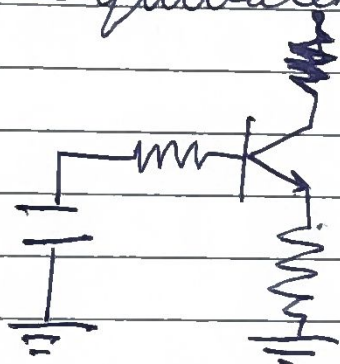
Calculation:-



(A)

DC Analysis

D.C. Equivalent Circuit, Thevenin Equivalent circuit



KVL on input side of circuit

$$I_B = \frac{V_{th} - V_{BE}}{R_{th} + (\beta + 1)R_E} = \frac{2.54 - 0.7}{(1.73)(10^3) + (271)(10^2)}$$

Here,

Thus,

$$I_C = \beta I_B = (271)(6.75)(10^{-6}) = 1.82 \text{ mA}$$

$$V_{th} = \frac{V_{CC} R_2}{R_1 + R_2} = 2.54 \text{ V}$$

$$R_{th} = \frac{R_1 R_2}{R_1 + R_2} = 1.73 \text{ k}\Omega$$

We are Given

$$C_u = 1.5 \text{ pF}, \beta = 271, C_\pi = 11 \text{ pF}$$

(B)

Small Signal Parameter

At Room Temperature, $V_T = 26 \text{ mV}$

$$V_{gm} = \frac{I_C}{V_T} = \frac{(1.82)(10^{-3})}{26 \times 10^{-3}} = 70 \text{ mV/V}$$

(PTD)

$$r_{\pi} = \frac{\beta}{g_m} = \frac{386}{0.001} = 386 \text{ k}\Omega \quad 3.86 \text{ k}\Omega$$

$$r_{o,c} = \frac{V_T}{I_c} = 14.3 \Omega \quad V_o = \infty$$

$$r_o = \infty$$

(C) Mid frequency Voltage Gain

$\therefore R_E$ is bypassed by C_S in 3 conditions

$$A_v = -g_m r_{\pi} = (-70)(10^{-3})(3)(10^3)$$

$$[A_v]_{\text{mid}} = -210$$

(D) Low frequency Analysis

$$f_{LCC1} = \frac{1}{2\pi [R_{Si} + (R_E || r_{\pi})] C_{C1}}$$

$$f_{LCC2} = \frac{1}{2\pi [R_c + R_L] C_{C2}}$$

$$f_{LCE} = \frac{1}{2\pi [R_E' || (\frac{R_{Si} || R_i || R_o}{1+\beta} + r_{\pi})] C_E}$$

$$f_{LCC1} = 133 \text{ Hz}$$

$$f_{LCC2} = 33.03 \text{ Hz} \quad (R_{Si} \approx 0 \text{ in } || \text{ combination})$$

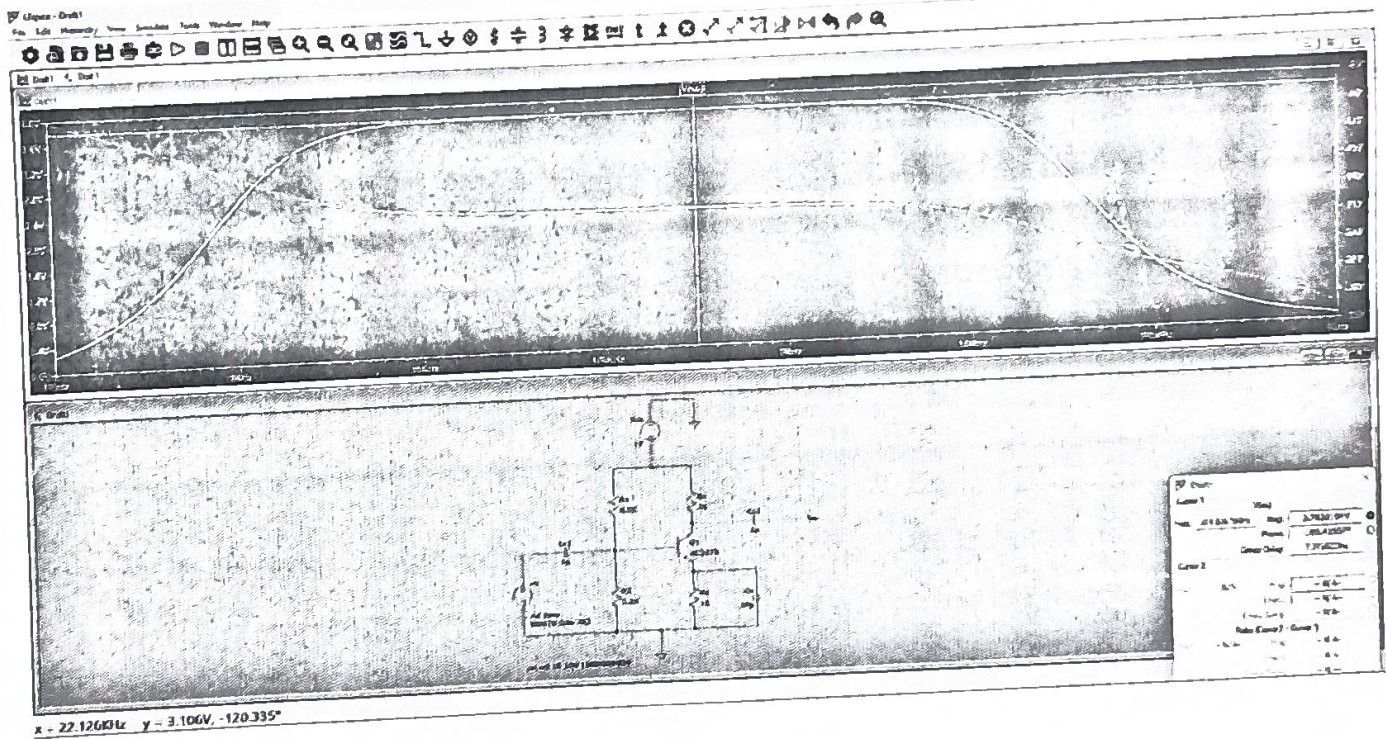
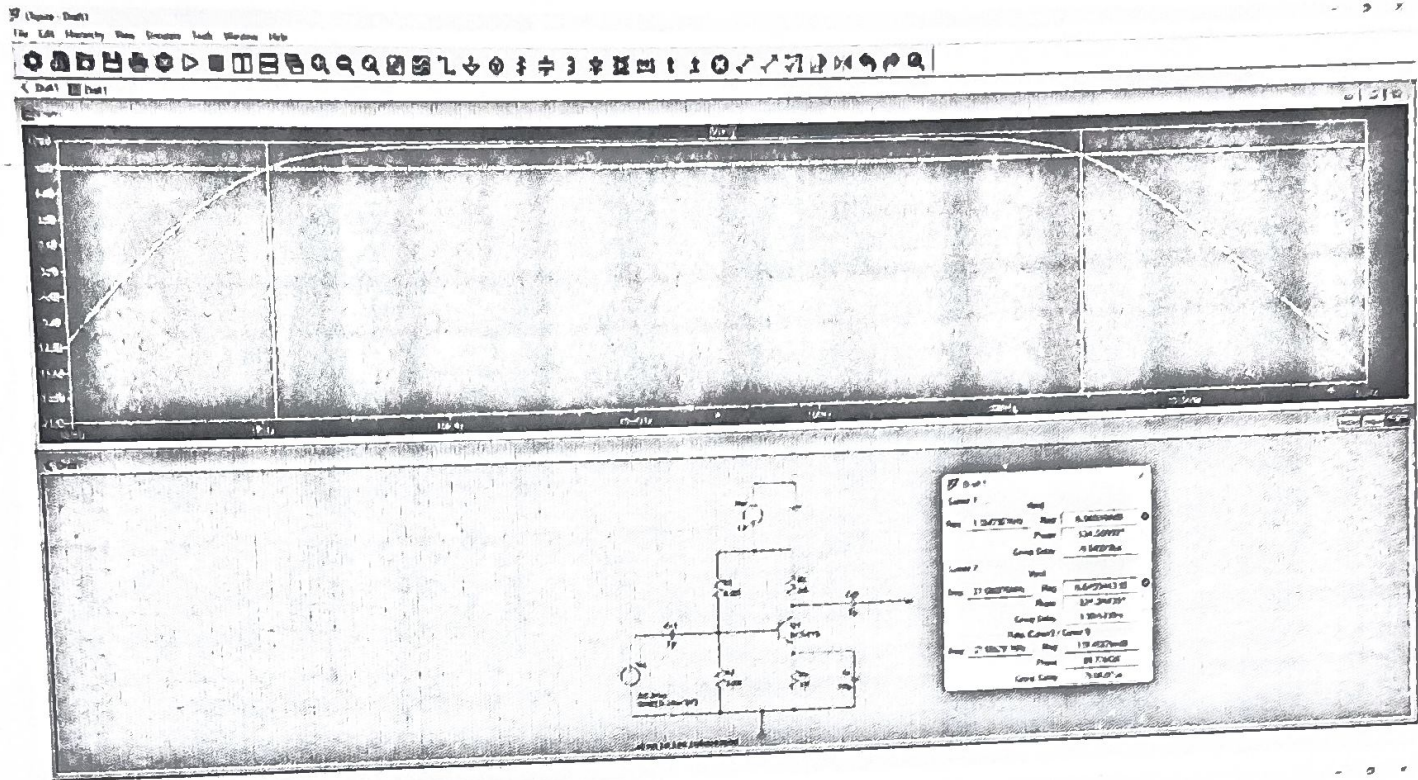
$$f_{LCE} = 1.14 \text{ kHz}$$

(E) High frequency Analysis :-

High frequency due to Miller's Effect

$$f_{H0} = \frac{1}{2\pi [R_{Si} || R_i || R_o || r_{\pi}]} = 35.4 \text{ MHz}$$

(F) Band Width = $f_2(\text{min}) - f_1(\text{max}) = 0.4453 \text{ MHz}$

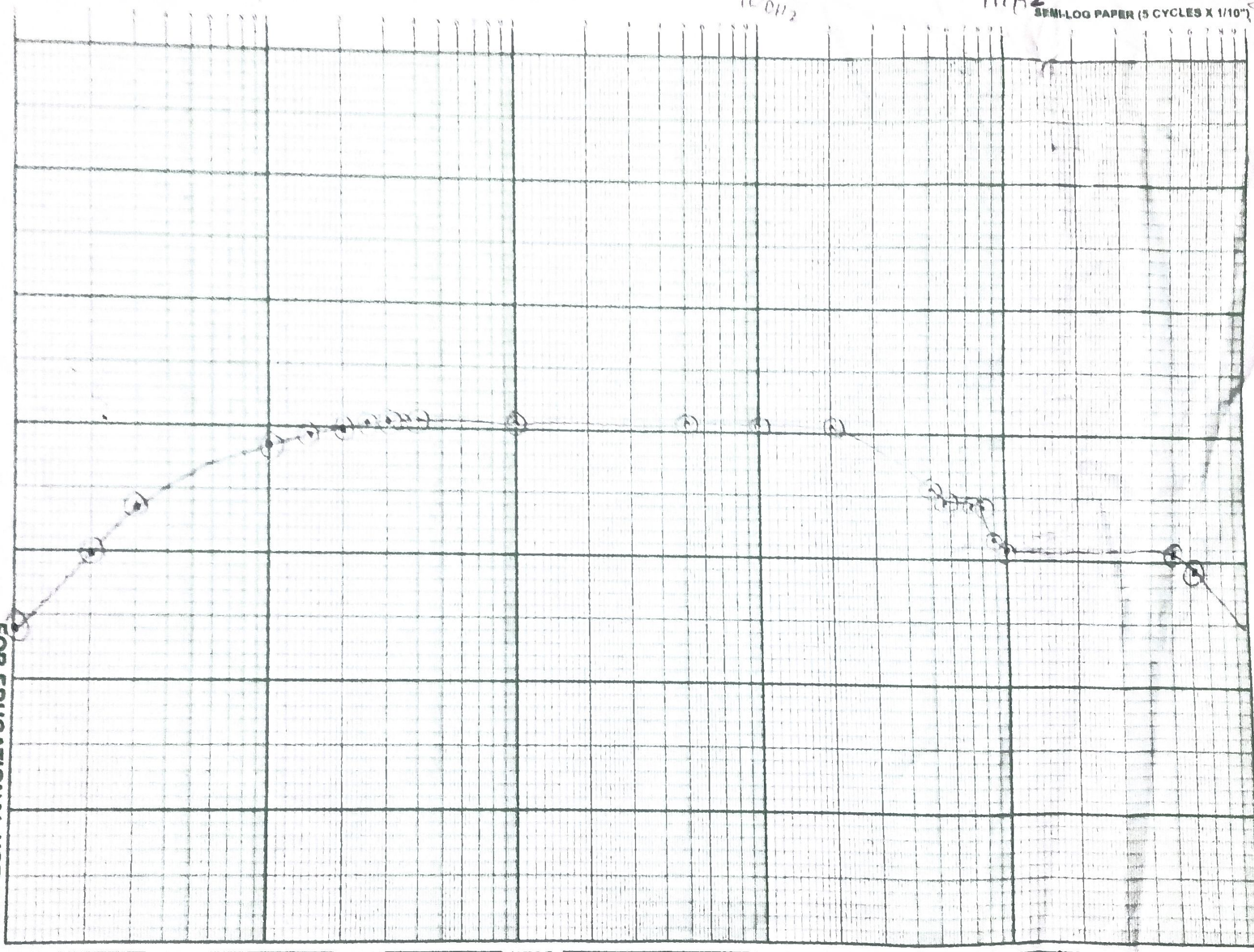


SEMI-LOG PAPER (5 CYCLES X 1/10")

100Hz 1KHz 10KHz 100KHz 1MHz 10MHz

70
60
50
40
30
20
10

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A PRODUCT



Name: DHAIRY KHAN 100Hz 1KHz 10KHz 100KHz 1MHz 10MHz

1001082404

1MHz

10MHz